

# Savit Pawar

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## EDUCATION

### University of Washington

Seattle, WA

Bachelor of Science in Computer Science - 3.91 GPA

Expected June 2028

- Relevant Coursework: Calculus III, Discrete Mathematics, Hardware/Software Interface, Data Structures and Parallelism, Linear Algebra, Software Engineering, Systems Programming

## EXPERIENCE

### Undergraduate Researcher

June 2026 – Present

PLSE (Programming Languages and Software Engineering Lab) at the University of Washington

Seattle, WA

- Built an **AI agent pipeline** that rewrites C code into memory-safe Rust, matching C output on **90%** of **7** tests.
- Made every translation automatically verifiable by porting C test suites to **Google Test**.
- Measured translation accuracy on real-world code by curating 2 open-source C repositories as validation benchmarks.
- Created an **LLVM-based C call-graph tool** to detect indirect function-pointer call sites missed by the standard pass.

### Teaching Assistant

June 2026 – Present

Paul G. Allen School of Computer Science

Seattle, WA

- Reinforced Hardware/Software Interface concepts for a **40+** student course by leading weekly quiz sections and discussions.

### Lab Systems Intern

Nov. 2025 – June 2026

SEAL (Sensors, Energy, and Automation Lab) at the University of Washington

Seattle, WA

- Cut per-person weekly reporting from **45 to 10 min** across a **50-person lab** by leading a Claude Code chatbot build.
- Drove a measured **37%** efficiency gain by maintaining two lab-wide tracking systems for lab coordination.
- Improved workload visibility by redesigning the lab dashboard to surface each member's completed work and next tasks.

## PROJECTS

### Transit Reliability | Python, PostgreSQL, Docker, FastAPI, Redpanda, GTFIS

- Built a real-time transit data pipeline tracking **350+** Seattle buses every 30 s, with **97%** matched to the schedule.
- Made the API **37×** faster (1.1 s to 0.03 s) by keeping track of mean instead of median due to the less data required.
- Created **4** Docker services for data processing and deployed **5** more open source services to production.

### Sokoban Environment | Python, JavaScript, BFS pathfinding, JSON/HTTP API, Three.js

- Won track and placed 2nd overall out of all teams at a **Google-sponsored, Google-judged** hackathon.
- Built a Sokoban RL benchmark for **LLM agent 3D perception** with elevation/box mechanics and BFS-verified maps.
- Visualized agent thought process move-by-move in a 2D/3D browser replay viewer to better understand decision making.

### LaunchHQ | Next.js, Vercel, Neon, Stripe

- Grew a full-stack SaaS waitlist platform to **40+ users** and **20+ live waitlists** via billing, custom domains, and widgets.
- Reduced signup friction with an embeddable widget and referral system that collects signups on client sites.
- Cut email setup to **near-zero config** by integrating Resend for transactional email and Stripe webhooks for billing.

### Talus | Python, XGBoost, scikit-learn, SHAP, Google Earth Engine

- Achieved **0.90 AUC-ROC** predicting landslide susceptibility across Washington at 10m via XGBoost and spatial CV.
- Flagged **46.8%** of landslide pixels as High/Very High risk (vs **15%** stable) on a blind holdout of the 2014 Oso zone.
- Predicted across **450M pixels** from 8 features over 7 geospatial sources; SHAP ranked precipitation and elevation top.
- Designed chunked raster pipeline for statewide predictions on **24GB RAM**, processing **60K×40K** grids in 1000-row strips.

## TECHNICAL SKILLS

**Languages:** Java, Python, JavaScript, HTML/CSS, SQL, Rust, C

**Technologies:** React, Next.js, Tailwind CSS, Streamlit, Claude Code, prompt engineering, XGBoost, scikit-learn, SHAP, pandas, NumPy, Function Calling, Hugging Face, GitHub Copilot, Git, Power BI, VS Code, Google Test, Docker, Redis, Redpanda, FastAPI

**Concepts:** Databases, Data Structures, Software Engineering, CI/CD, Caching, Parallelism, APIs, Version Control, Containerization